WORKED SOLUTION

STEP II 1999, Question 2

Pure • Discriminants of Quadratics

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q2

The key idea

Everything hinges on the discriminant of $(*)$. Because the x-coefficient carries a factor of 2, the quarter-discriminant $\frac{1}{4}(b^2 - 4ac)$ is especially tidy, and it turns out to be a quadratic in n. So each part becomes a question about the sign of a *second* quadratic, this time with n as the variable.

Solution

Throughout we take $pn^2 + q \geq 0$, so that the square root is real and $(*)$ is an ordinary quadratic with real coefficients. (This holds automatically in parts (i) and (iii), where $q > 0$ or $q \geq 0$.) With $a = n$, $b = 2\sqrt{pn^2 + q}$ and $c = rn + s$, the equation has real roots exactly when $b^2 - 4ac \geq 0$, that is, when $\frac{1}{4}b^2 - ac \geq 0$. Now

$$\begin{aligned} \tfrac{1}{4}b^2 - ac &= (pn^2 + q) - n(rn + s) \\ &= (p - r)n^2 - sn + q. \end{aligned}$$

Write $f(n) = (p - r)n^2 - sn + q$. Then

$$(*) \text{ has no real roots} \iff f(n) < 0.$$

Part (i)

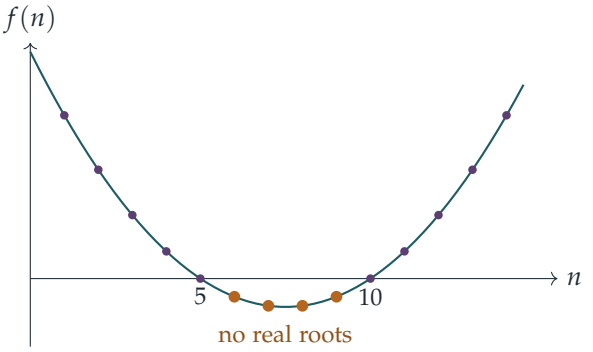
Here $p - r = 1$, so

$$f(n) = n^2 - 15n + 50 = (n - 5)(n - 10).$$

This is negative exactly when $5 < n < 10$. Since n is a positive integer, $(*)$ has no real roots precisely for

$$n \in \{6, 7, 8, 9\}.$$

At $n = 5$ and $n = 10$ we have $f(n) = 0$, so there is a repeated real root, and those values are excluded.



Part (ii)

Now $p < r$, so the leading coefficient $p - r$ of f is negative. Think of f as a quadratic in a real variable t:

$$f(t) = (p-r)t^2 - st + q.$$

Its discriminant is

$$s^2 - 4(p-r)q,$$

which is negative by the hypothesis $4q(p-r) > s^2$.

Ask yourself Why does a negative discriminant for f tell us anything about the sign of $f(n)$?

A quadratic with negative discriminant has no real zeros, so its graph never crosses the axis and it keeps one sign for all real t. With a negative leading coefficient the graph is an upside-down parabola, so that sign is negative. Explicitly, completing the square,

$$f(t) = (p-r)\Big(t - \frac{s}{2(p-r)}\Big)^2 + \frac{4q(p-r) - s^2}{4(p-r)},$$

and both terms are negative or zero: the first because $p - r < 0$, and the second because its numerator is positive and its denominator negative. The second term is strictly negative, so $f(t) < 0$ for every real t.

In particular $f(n) < 0$ for every positive integer n, so $(*)$ has no real roots for any value of n, as required.

Note that the hypotheses force $q < 0$: since $p - r < 0$ and $4q(p-r) > s^2 \geq 0$, q must be negative. So $pn^2 + q$ really can be negative for small n, and our standing assumption that the square root is real matters here. (If an imaginary coefficient were allowed, $x = 0$ could be a root: try $p = 1$, $r = 2$, $s = -2$, $q = -10$, $n = 1$.)

Part (iii)

Now $n = 1$, $p - r = 1$ and $q = s^2/8 \geq 0$, so $p + q > 0$ and the coefficients are real. Here

$$f(1) = 1 - s + \frac{s^2}{8} = \frac{s^2 - 8s + 8}{8}.$$

Real roots occur if, and only if, $f(1) \geq 0$, that is, if, and only if,

$$s^2 - 8s + 8 \geq 0.$$

The roots of $s^2 - 8s + 8 = 0$ are

$$s = \frac{8 \pm \sqrt{64 - 32}}{2} = 4 \pm 2\sqrt{2},$$

so $s^2 - 8s + 8 = \big(s - (4 - 2\sqrt{2})\big)\big(s - (4 + 2\sqrt{2})\big)$. This product is non-negative exactly when s lies outside the open interval between the roots. Hence $(*)$ has real roots if, and only if,

$$s \leq 4 - 2\sqrt{2} \quad \text{or} \quad s \geq 4 + 2\sqrt{2},$$

as required. Every step is reversible, which is what gives us both directions of the "if and only if". At the two endpoints the root is repeated.

Checks

- In (i), take $n = 7$: $\frac{1}{4}b^2 = 3 \cdot 49 + 50 = 197$ and $ac = 7 \cdot 29 = 203$, so $\frac{1}{4}b^2 - ac = -6 < 0$. Take $n = 11$: $413 - 11 \cdot 37 = 6 > 0$.
- In (iii), $4 - 2\sqrt{2} \approx 1.1716$ and $4 + 2\sqrt{2} \approx 6.8284$. With $s = 4$ we get $f(1) = 1 - 4 + 2 = -1 < 0$, which is inside the gap, as expected.
- A computer searched n from 1 to 199 for (i), tested about 5 million random cases of (ii), and tested 10^5 random values of s for (iii). All agreed with the results above.

Common slips

- Including $n = 5$ and $n = 10$ in (i). There the discriminant is zero, so there is a (repeated) real root.
- In (ii), checking only that $f(n) < 0$ for large n. The point of the discriminant condition is that f is negative *everywhere*, including small n.
- Forgetting the factor of 2 in the x-coefficient and getting $4(pn^2 + q)$ in the wrong place in the discriminant.

Beyond STEP

The move here, treating a parameter as a new variable and asking about the sign of a quadratic in it, is the start of studying how the roots of a polynomial depend on its coefficients. At university this leads to discriminants of higher-degree polynomials, and to the idea that a quadratic form $at^2 + bt + c$ is negative definite exactly when $a < 0$ and $b^2 < 4ac$. That same test, for two variables, classifies stationary points of functions of several variables. Siklos's *Advanced Problems in Mathematics* has plenty more STEP questions built on careful discriminant arguments.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.